Hyperuniformity of One-Dimensional Point Patterns: A Complete Project Summary

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March 2026

Abstract

This document provides a comprehensive introduction to the concept of hyperuniformity and summarizes the results of a two-week research project analyzing hyperuniform point patterns in one dimension. Starting from basic principles accessible to newcomers, we explain what hyperuniformity means, why it matters, and how it is measured. We then present our analysis of 26 different point patterns spanning all three hyperuniformity classes, including the novel result of finding a quasicrystal with hyperuniformity exponent $\alpha = 1.545$, which fills a previously unexplored gap in the hyperuniformity spectrum.

Contents

1 Introduction: What is Hyperuniformity?

1.1 The Basic Idea

Imagine you're looking at a box of cereal. If you scoop out a cup from different parts of the box, you'll get roughly the same amount each time—but not exactly the same. There will be small random variations. These variations in "how much stuff is in a region" are called **density fluctuations**.

Now imagine three different arrangements of points on a line:

1. **Random (Poisson)**: Points placed completely at random, like raindrops on a sidewalk. If you count how many points are in a window, the count varies a lot from window to window.
2. **Regular lattice**: Points spaced perfectly evenly, like fence posts. The count in any window is almost exactly predictable.
3. **Hyperuniform**: Something in between—the points aren't perfectly regular, but the density fluctuations are *suppressed* compared to random placement.

Hyperuniformity is a property of point patterns (or more generally, any distribution of matter) where large-scale density fluctuations are anomalously suppressed. The pattern looks "locally disordered" but is "globally ordered" in a statistical sense.

1.2 Why Does This Matter?

Hyperuniform materials have unusual physical properties:

- **Photonic applications**: Hyperuniform structures can have photonic band gaps (blocking certain wavelengths of light) even without periodic order.
- **Materials science**: Understanding hyperuniformity helps design materials with specific optical, mechanical, or transport properties.
- **Biology**: The arrangement of photoreceptors in bird eyes and the distribution of cells in certain tissues are hyperuniform.
- **Cosmology**: The distribution of galaxies in the universe is approximately hyperuniform on large scales.

1.3 A Precise Definition

For those comfortable with mathematics, here's the precise definition:

A point pattern is **hyperuniform** if its structure factor $S(k)$ vanishes as the wavenumber k approaches zero:

$$\lim_{k \to 0} S(k) = 0$$

The structure factor $S(k)$ measures the amplitude of density fluctuations at wavelength $2\pi/k$. For a random (Poisson) pattern, $S(k) = 1$ for all k. For a hyperuniform pattern, long-wavelength fluctuations are suppressed.

Don't worry if this seems abstract—we'll explain what this means in practical terms throughout this document.

2 Measuring Hyperuniformity

2.1 Number Variance: Counting Points in Windows

The most intuitive way to measure hyperuniformity is through the **number variance** $\sigma^2(R)$.

The procedure:

1. Take your point pattern (a set of points on a line).
2. Place a "window" of half-width R at a random location.
3. Count how many points N fall inside the window.
4. Repeat many times at different random locations.
5. Compute the variance (spread) of these counts.

For a **random (Poisson)** pattern, the variance grows linearly with window size:

$$\sigma^2_{\text{Poisson}}(R) = 2\rho R$$

where ρ is the number density (average points per unit length).

For a **hyperuniform** pattern, the variance grows *slower* than linear—this is what "suppressed fluctuations" means mathematically.

2.2 The Three Classes of Hyperuniformity

Based on how the variance grows with window size, hyperuniform patterns fall into three classes:

Class	**Variance Growth**	**Exponent α**	**Physical Meaning**
Class I	Bounded (constant)	$\alpha > 1$	Strongest suppression
Class II	$\sim \ln R$ (logarithmic)	$\alpha = 1$	Intermediate
Class III	$\sim R^{1-\alpha}$ (power law)	$0 < \alpha < 1$	Weakest suppression
Non-hyperuniform	$\sim R$ (linear)	$\alpha \leq 0$	No suppression (random)

Table 1: The three classes of hyperuniformity, distinguished by how number variance grows with window size.

2.3 The Hyperuniformity Exponent α

The **hyperuniformity exponent** α quantifies how strongly fluctuations are suppressed. It's defined by how the structure factor vanishes near $k = 0$:

$$S(k) \sim |k|^\alpha \quad \text{as } k \to 0$$

- Higher α = fluctuations suppressed more strongly
- $\alpha > 1$: Class I (bounded variance)
- $\alpha = 1$: Class II (logarithmic variance)
- $0 < \alpha < 1$: Class III (power-law variance)
- $\alpha = 0$: Not hyperuniform

For a perfect lattice (evenly spaced points), $\alpha = \infty$—the ultimate hyperuniform pattern.

2.4 The Surface-Area Coefficient $\bar{\Lambda}$

For Class I patterns (where variance is bounded), we can define a useful quantity called the **surface-area coefficient**:

$$\bar{\Lambda} = \lim_{R\to\infty} \frac{\sigma^2(R)}{2R}$$

This measures the "local disorder" of the pattern:

- Lower $\bar{\Lambda}$ = more ordered (less local fluctuation)
- Perfect lattice: $\bar{\Lambda} = 1/6 \approx 0.167$ (the minimum possible)
- Higher values indicate more disorder while still being hyperuniform

For Classes II and III, the variance grows without bound, so $\bar{\Lambda}$ diverges (doesn't exist as a finite number).

3 Quasicrystals and Substitution Tilings

3.1 What is a Quasicrystal?

A **crystal** is a perfectly periodic arrangement of atoms—the same pattern repeats over and over. A **quasicrystal** is ordered but *not* periodic. It has long-range order (you can predict where atoms will be far away) but never exactly repeats.

The classic example is the Penrose tiling—a pattern of tiles that fills the plane without gaps or overlaps, following specific rules, but never repeating periodically.

Quasicrystals were discovered experimentally in 1984 by Dan Shechtman (who won the Nobel Prize for this in 2011). They exhibit "forbidden" symmetries that are impossible for periodic crystals.

3.2 Building 1D Quasicrystals with Substitution Rules

In one dimension, we can build quasicrystals using **substitution rules**. Here's how:

1. Start with an alphabet of tile types (e.g., Short "S" and Long "L").
2. Define rules for replacing each tile with a sequence of tiles.
3. Apply the rules repeatedly to grow the pattern.
4. Place points at the boundaries between tiles.

Example: The Fibonacci Chain
Rules: $S \to L$, $L \to LS$
Starting with L:

$$\begin{aligned}
\text{Step 0:} &\quad L \\
\text{Step 1:} &\quad LS \\
\text{Step 2:} &\quad LSL \\
\text{Step 3:} &\quad LSLLS \\
\text{Step 4:} &\quad LSLLSLSL \\
&\quad \vdots
\end{aligned}$$

The ratio of L's to S's approaches the golden ratio $\phi = (1 + \sqrt{5})/2 \approx 1.618$.

To get a point pattern, we assign lengths to the tiles (e.g., $\ell_L = \phi$, $\ell_S = 1$) and place points at each tile boundary.

3.3 The Substitution Matrix

The growth of the pattern is governed by a **substitution matrix** M. For a two-letter alphabet $\{S, L\}$, the matrix entry M_{ij} counts how many tiles of type i appear when tile j is substituted.

For Fibonacci: $S \to L$ (one L, zero S's), $L \to LS$ (one L, one S)

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

The eigenvalues of this matrix determine the pattern's properties:

- The largest eigenvalue λ_1 gives the growth rate.
- The ratio of tile lengths comes from the eigenvector.
- The hyperuniformity exponent α depends on the eigenvalue ratio.

3.4 The Eigenvalue Formula for α

For substitution tilings, there's an elegant formula relating the hyperuniformity exponent to the substitution matrix eigenvalues:

$$\boxed{\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln |\lambda_1|}}$$

where λ_1 is the largest eigenvalue (in absolute value) and λ_2 is the second largest.

This formula is powerful: once you know the substitution matrix, you can immediately predict α without doing any simulations!

4 Project Goals

The goal of this research project was to:

1. **Implement** numerical algorithms to analyze hyperuniform point patterns in 1D.
2. **Validate** the code against known analytical results.
3. **Analyze** a variety of patterns from all three hyperuniformity classes.
4. **Rank** patterns by their degree of hyperuniformity using α and $\bar{\Lambda}$.
5. **Fill gaps** in the hyperuniformity "spectrum" by finding patterns with intermediate α values.

The project was conducted over two weeks under the guidance of Professor Salvatore Torquato at Princeton University.

5 Week 1 Results: Metallic-Mean Quasicrystals

5.1 The Three Metallic Means

In Week 1, we analyzed three quasicrystal chains based on "metallic means"—generalizations of the golden ratio:

Pattern	**Substitution Rule**	**Metallic Mean**	**Value**
Fibonacci	$S \to L$, $L \to LS$	Golden	$\phi = 1.618$
Silver	$S \to L$, $L \to LLS$	Silver	$\mu_2 = 2.414$
Bronze	$S \to L$, $L \to LLLS$	Bronze	$\mu_3 = 3.303$

Table 2: The three metallic-mean substitution chains analyzed in Week 1.

5.2 Validation: The Poisson Benchmark

Before analyzing quasicrystals, we validated our variance computation code using a Poisson (random) pattern, where we know the exact answer: $\sigma^2(R) = 2\rho R$.

Our numerical result matched the theory to within **1.1% error**, confirming the code was working correctly.

5.3 Results: All Three Give $\alpha = 3$

Surprisingly, all three metallic-mean chains gave the same hyperuniformity exponent:

Pattern	$\boldsymbol{\alpha}$ (measured)	$\boldsymbol{\alpha}$ (expected)	$\bar{\boldsymbol{\Lambda}}$
Fibonacci	3.05	3	0.200
Silver	2.99	3	0.250
Bronze	2.99	3	0.282

Table 3: Week 1 results for metallic-mean chains.

Why is $\alpha = 3$ for all of them?

For 2×2 metallic-mean matrices with determinant ±1, there's a mathematical constraint:

$$|\lambda_2| = \frac{1}{|\lambda_1|}$$

Plugging into the eigenvalue formula:

$$\alpha = 1 - \frac{2\ln(1/|\lambda_1|)}{\ln|\lambda_1|} = 1 - \frac{-2\ln|\lambda_1|}{\ln|\lambda_1|} = 1 + 2 = 3$$

No matter which metallic mean we choose, we always get $\alpha = 3$!

5.4 Week 1 Conclusions

- Validated numerical code against Poisson benchmark.
- Confirmed $\alpha = 3$ for all metallic-mean chains (Class I).

- Observed that $\bar{\Lambda}$ varies between patterns even when α is the same.
- Identified a limitation: 2×2 matrices can only give $\alpha = 3$.

6 Week 2 Results: Expanding the Spectrum

6.1 Goal: Cover All Three Hyperuniformity Classes

After Week 1, our ranking only included Class I patterns with $\alpha = 3$. Week 2 aimed to:

1. Find Class II ($\alpha = 1$) and Class III ($\alpha < 1$) patterns.
2. Analyze non-quasicrystal hyperuniform patterns (perturbed lattices, stealthy).
3. Fill the gap: find a quasicrystal with $1 < \alpha < 2$.

6.2 Class II: The Period-Doubling Chain

The Period-Doubling chain has the substitution rule:

$$S \to L, \quad L \to LSS$$

with matrix:

$$M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}, \quad \lambda_1 = 2, \quad |\lambda_2| = 1$$

Since $|\lambda_2| = 1$, we get $\ln(1) = 0$, so:

$$\alpha = 1 - \frac{2 \times 0}{\ln 2} = 1$$

Results:

- Variance grows logarithmically: $\sigma^2(R) \sim \ln R$
- $\bar{\Lambda}$ diverges (no finite limit)
- **Class II confirmed**

6.3 Class III: The 0222 Chain

The 0222 chain has the substitution rule:

$$S \to LL, \quad L \to SSLL$$

with matrix:

$$M = \begin{pmatrix} 0 & 2 \\ 2 & 2 \end{pmatrix}, \quad \lambda_1 = 1 + \sqrt{5}, \quad |\lambda_2| = \sqrt{5} - 1 \approx 1.236$$

Since $|\lambda_2| > 1$, we get $\ln|\lambda_2| > 0$, so:

$$\alpha = 1 - \frac{2\ln(\sqrt{5} - 1)}{\ln(1 + \sqrt{5})} \approx 0.64$$

Results:

- Variance grows as power law: $\sigma^2(R) \sim R^{0.36}$
- $\bar{\Lambda}$ diverges
- **Class III confirmed**

6.4 Perturbed Lattices: The URL

The **Uncorrelated Random Lattice (URL)** is constructed by:

1. Starting with a perfect integer lattice (points at $0, 1, 2, 3, \ldots$).
2. Displacing each point by a random amount $\delta \in [-a, a]$.
3. The displacements are independent (uncorrelated).

The URL has an exact analytical formula:

$$\bar{\Lambda}(a) = \frac{1}{6} + \frac{a^2}{3}$$

and always has $\alpha = 2$ regardless of a.

Numerical verification:

a	$\bar{\Lambda}$ (numerical)	$\bar{\Lambda}$ (exact)
0.1	0.168	0.170
0.5	0.208	0.208
1.0	0.334	0.333

Table 4: URL validation: numerical results match theory to $< 0.3\%$.

This excellent agreement validates our numerical code against an independent analytical result.

6.5 Stealthy Hyperuniform Patterns

A **stealthy** pattern is hyperuniform with the additional property that $S(k) = 0$ for all $|k| < K^*$, not just at $k = 0$. These patterns are "invisible" to probes at long wavelengths.

We analyzed stealthy configurations provided by another group member. These were generated using **collective coordinate optimization**:

1. Start with random point positions.
2. Define an energy function based on $S(k)$ in the exclusion region.
3. Use gradient descent to minimize the energy.
4. The points rearrange themselves to make $S(k) = 0$ for $|k| < K^*$.

Results:

χ	$\bar{\Lambda}$	α	Iterations
0.1	0.166	2.0	328
0.2	0.166	2.0	596
0.3	0.166	2.0	694

Table 5: Stealthy pattern results. The parameter $\chi = K^*/(4\pi\rho)$ controls the size of the exclusion region.

Remarkably, $\bar{\Lambda} \approx 0.166 \approx 1/6$—the same as a **perfect lattice**! Stealthy patterns achieve optimal hyperuniformity despite being disordered.

6.6 The Gap Problem: $1 < \alpha < 2$

After analyzing all these patterns, we noticed a gap in our ranking:

α value	Class	Examples
$\alpha = 3$	I	Fibonacci, Silver, Bronze
$\alpha = 2$	I	URL, Stealthy
$1 < \alpha < 2$	I	??? No quasicrystal known!
$\alpha = 1$	II	Period-Doubling
$\alpha < 1$	III	0222 Chain

Table 6: The gap in the hyperuniformity spectrum before the Bombieri-Taylor analysis.

The problem: All our quasicrystals came from 2×2 substitution matrices, which are constrained to give either $\alpha = 3$ (metallic means), $\alpha = 1$ (period-doubling), or $\alpha < 1$.

The solution: Use a 3×3 matrix (cubic irrational) to break the constraint!

6.7 The Bombieri-Taylor Cubic Substitution

Professor Torquato suggested looking at a cubic substitution from Bombieri & Taylor's 1986 paper on quasicrystal diffraction.

The substitution rule (3-letter alphabet):

$$a \to aac, \quad b \to ac, \quad c \to b$$

The substitution matrix:

$$M = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Characteristic equation: $x^3 - 2x^2 - x + 1 = 0$

Eigenvalues:

- $\theta_1 = 2.247$ (largest, a Pisot number)
- $|\lambda_2| = 0.802$
- $|\lambda_3| = 0.555$

Note that $|\lambda_2| = 0.802 \neq 1/\theta_1 = 0.445$—the 3×3 matrix breaks the constraint!

Predicted exponent:

$$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = 1 - \frac{2(-0.221)}{0.809} = \boxed{1.545}$$

6.8 Implementation and Results

We extended our code to handle 3-letter alphabets:

- Tile lengths from eigenvector: $\ell_a = 4.049$, $\ell_b = 2.247$, $\ell_c = 1.000$
- Generated pattern with 25 iterations: $N = 9.5$ million points

- Domain length: $L = 22.9$ million

Results:

- $\sigma^2(R)$ is **bounded** $\Rightarrow$ Class I confirmed
- $\bar{\Lambda} = 0.377$
- $\alpha = 1.545$ (from eigenvalue formula)
- **Gap filled!**

The Bombieri-Taylor cubic substitution is the first known quasicrystal with $1 < \alpha < 2$.

7 Complete Hyperuniformity Ranking

After two weeks of analysis, we have characterized 26 different point patterns:

Pattern	**Class**	$\boldsymbol{\alpha}$	$\bar{\boldsymbol{\Lambda}}$
Integer Lattice	I	∞	0.167
Stealthy ($\chi = 0.1$–0.3)	I	2.0	0.166
URL ($a = 0.1$)	I	2.0	0.168
Fibonacci	I	3.0	0.200
URL ($a = 0.5$)	I	2.0	0.208
Silver	I	3.0	0.250
Bronze	I	3.0	0.282
URL ($a = 1.0$)	I	2.0	0.333
Bombieri-Taylor	**I**	**1.545**	**0.377**
Period-Doubling	II	1.0	diverges
0222 Chain	III	0.64	diverges

Table 7: Complete hyperuniformity ranking of analyzed patterns, sorted by $\bar{\Lambda}$ within each class.

Key observations:

1. The perfect lattice and stealthy patterns achieve the optimal $\bar{\Lambda} \approx 1/6$.
2. Different α values don't necessarily correspond to different $\bar{\Lambda}$—the two metrics capture different aspects of hyperuniformity.
3. The Bombieri-Taylor cubic fills the gap between $\alpha = 1$ and $\alpha = 2$.
4. Classes II and III have divergent $\bar{\Lambda}$, so they can only be compared using α.

8 Methods Summary

8.1 Pattern Generation

- **Substitution tilings**: Iteratively apply substitution rules, compute tile lengths from eigenvector, place points at tile boundaries.

- **URL**: Generate lattice points, add uniform random displacements.
- **Stealthy**: Start random, optimize positions to minimize $S(k)$ in exclusion region.

8.2 Number Variance Computation

1. For each window size R, place many random windows.
2. Count points in each window using binary search (efficient for large patterns).
3. Compute variance of counts.
4. Extract $\bar{\Lambda}$ from large-R behavior.

8.3 Extracting α

Two approaches:

1. **Eigenvalue formula**: Direct calculation from substitution matrix (exact for quasicrystals).
2. **Spreadability method**: Compute excess spreadability $E(t)$ from structure factor, fit $E(t) \sim t^{-(1+\alpha)/2}$ at long times.

For quasicrystals, the eigenvalue formula is preferred because log-periodic oscillations make fitting difficult.

9 Key Findings and Significance

9.1 Main Results

1. **Validated numerical code** against Poisson (1.1% error) and URL ($< 0.3\%$ error) benchmarks.
2. **Confirmed eigenvalue formula** for α in substitution tilings across 2×2 and 3×3 matrices.
3. **Covered all three hyperuniformity classes**:
   - Class I: Metallic means, URL, Stealthy, Bombieri-Taylor
   - Class II: Period-Doubling
   - Class III: 0222 Chain
4. **Discovered first quasicrystal with** $1 < \alpha < 2$: The Bombieri-Taylor cubic substitution with $\alpha = 1.545$.
5. **Demonstrated** that stealthy patterns achieve lattice-level order ($\bar{\Lambda} \approx 1/6$) despite being disordered.

9.2 Scientific Significance

The Bombieri-Taylor result is significant because:

- It shows that cubic irrational substitutions can access α values impossible for quadratic metallic means.
- It fills a gap in the hyperuniformity "spectrum," providing a new data point for understanding the relationship between substitution structure and hyperuniformity.
- It validates the eigenvalue formula for 3×3 matrices, extending its known applicability.

9.3 Open Questions

1. Can we find other cubic or quartic substitutions with different α values in the range $(1, 2)$?
2. How do these results generalize to 2D and 3D quasicrystals?
3. What physical properties distinguish patterns with different α values?
4. Is there a systematic way to construct substitutions with any desired α?

10 Glossary of Terms

Hyperuniformity Property of a point pattern where large-scale density fluctuations are suppressed; $S(k) \to 0$ as $k \to 0$.

Structure Factor $S(k)$ Fourier transform of density-density correlations; measures fluctuation amplitude at wavenumber k.

Number Variance $\sigma^2(R)$ Variance of point counts in windows of half-width R.

Hyperuniformity Exponent α Rate at which $S(k) \sim |k|^\alpha$ vanishes; higher α = stronger suppression.

Surface-Area Coefficient $\bar{\Lambda}$ For Class I patterns, the limiting ratio $\sigma^2(R)/(2R)$; measures local disorder.

Quasicrystal Ordered but non-periodic arrangement of matter; has long-range order without translational periodicity.

Substitution Tiling Pattern generated by repeatedly applying replacement rules to tiles.

Metallic Mean Generalization of golden ratio; solutions to $x^2 = nx + 1$.

Pisot Number Algebraic integer > 1 whose conjugate roots all have absolute value < 1.

Stealthy Pattern Hyperuniform pattern with $S(k) = 0$ for all $|k| < K^*$, not just at $k = 0$.

URL (Uncorrelated Random Lattice) Lattice with independent random displacements at each site.

Spreadability Diffusion-based method for probing microstructure; the excess spreadability $E(t)$ decays as $t^{-(1+\alpha)/2}$.

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Acknowledgments

This research was conducted under the guidance of Professor Salvatore Torquato at Princeton University. Special thanks for suggesting the Bombieri-Taylor cubic substitution as a way to fill the $1 < \alpha < 2$ gap.